

No-Code Statistical Inference:

Understanding Data, Uncertainty, and Evidence

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Abstract

Statistical inference is often introduced through a sequence of computational procedures: calculate a descriptive statistic, select a test, obtain a p -value, fit a model, and report output. This sequence can obscure the prior and more consequential work of defining observational units, specifying a target population and estimand, understanding the sampling and measurement process, assessing assumptions, and interpreting uncertainty conditional on a model. This position paper advances a different pedagogical orientation. It treats statistical inference as a structured argument linking data-generating processes, formal probability models, estimators, diagnostics, and substantive conclusions. Throughout, a parameter is denoted by an unadorned Greek letter such as θ , its estimator by $\hat{\theta}$, and an estimated standard error by $\widehat{SE}(\hat{\theta})$ unless a method-specific notation is explicitly introduced. The paper develops the mathematical foundations of this argument across data and measurement, sampling and estimation, comparisons and categorical evidence, regression, model checking, prediction, and reproducibility. It also describes a guarded graphical learning environment in which programming is not a prerequisite for engaging with these ideas. The central claim is deliberately modest: reducing the need to write code can make statistical reasoning more accessible only if the interface keeps assumptions, diagnostics, uncertainty, effect sizes, and limitations visible rather than replacing them with automated decisions.

Keywords: statistical inference; estimands; uncertainty; confidence intervals; hypothesis testing; regression; diagnostics; reproducibility; statistics education.

1 Introduction

Statistical practice is not defined by the use of a particular formula, a particular programming language, or a particular software interface. It is defined by a chain of claims. An analyst begins with a scientific or practical question, identifies the population and the quantity about which a claim is sought, observes data produced by an imperfect sampling and measurement process, selects a model or procedure, quantifies uncertainty, examines the compatibility of assumptions with the data and design, and states a conclusion whose scope is limited by that chain. The numerical calculation is essential, but it is only one link in the inferential argument.

This perspective has two implications for statistical education. First, introductory instruction should not present techniques as a menu of tests detached from their data-generating conditions. A t -test, a contingency-table test, a linear regression coefficient, and an out-of-sample prediction score are answers to different formal questions. Their interpretation depends on the observational

unit, dependence structure, measurement scale, missing-data rule, target estimand, and model assumptions. Second, instruction should distinguish computational accessibility from inferential automation. It is possible to lower a programming barrier while increasing, rather than decreasing, attention to the reasoning that makes an analysis credible.

The paper is organized as a position paper rather than a methods manual. Section 2 clarifies the limited role of a graphical learning environment. The subsequent sections develop the statistical content itself: the definition of data and estimands; sampling, estimation, and uncertainty; evidence from comparisons and categorical tables; regression and model specification; and model checking, prediction, validation, and reproducibility. The argument throughout is that inference is best taught as a transparent sequence of conditional claims. Foundational treatments of statistical inference emphasize this connection between probability models, estimators, and evidential statements [3, 14]; exploratory analysis and modeling texts similarly emphasize that data inspection and model criticism are not optional preliminaries [7, 8, 12].

2 A guarded graphical learning environment

The accompanying Streamlit application is available at <https://no-code-statistical-inference.streamlit.app/>. It is intended as a guarded teaching and analysis environment, not as an unrestricted statistical decision engine. By “no-code,” we do not mean that statistical computation occurs without code. Computations are performed programmatically in the background, and the underlying code will be delivered with the seminar materials. Rather, participants are not required to write, read, or navigate R or Python code in order to work through the statistical material.

Participants engage with statistical concepts, research questions, assumptions, worked examples, visualizations, and interpretations through an interactive graphical interface. They may also upload CSV or Excel datasets manually and explore supported analyses without translating substantive questions into programming syntax. The interface is designed to begin with the research question and study structure, not with an undifferentiated list of tests. For supported workflows, it can expose compatible, cautionary, and incompatible analytical pathways; display data audits and missing-data choices; and make estimates, intervals, effect sizes, diagnostics, and limitations available for inspection.

This design does not eliminate the need for statistical judgment. It shifts attention away from software operation and toward the inferential choices that must remain the participant’s responsibility: defining the observational unit, identifying dependence, naming the estimand, selecting a missing-data rule, assessing the appropriateness of a model, interpreting uncertainty, and documenting limitations. The application should therefore be understood as a pedagogical scaffold. It can calculate only those procedures that have been implemented and guarded, and it cannot establish causal identification, certify a model as true, or select a method infallibly. Its educational purpose is to make the reasoning around calculation more visible while treating programming as an initial barrier that need not dominate the learning of statistics.

3 Data, Measurement, Exploratory Analysis, and the Estimand

Statistical inference begins before a model is selected or a test is calculated. It begins with a statement of what constitutes an observation, how variables represent features of those observations,

which population is being described, and which quantity is to be learned. These choices are not merely matters of data management. They determine the admissible comparisons, the meaning of uncertainty, and the scope of any conclusion. A numerical output can therefore be mathematically correct while answering a question that was never posed, or while attaching a population interpretation to measurements that do not support it. This section develops the statistical foundations on which later inferential claims depend: observational units and sampling, measurement and coding, descriptive and graphical analysis, missingness, and formal estimands.

3.1 Observational units, populations, and samples

An *observational unit* is the entity about which one or more measurements are recorded. Depending on the study, it may be a person, household, school, firm, medical visit, geographic region, transaction, or time interval. Let $i = 1, \dots, n$ index the recorded units, and let $\mathbf{X}_i = (X_{i1}, \dots, X_{ip})$ denote their measured attributes. The index i should not be confused with an independent replicate. Repeated observations from the same individual, members of the same family, patients treated by the same clinician, or measurements collected in the same cluster may be statistically dependent. The unit of analysis must consequently be specified together with the unit of observation and, when they differ, the level at which the scientific claim is made.

The *target population* is the collection of units to which the intended claim refers. The *sample* is the collection of units whose data are actually observed. A finite-population description concerns a fixed set $U = \{1, \dots, N\}$ and a sample $s \subset U$ selected according to a sampling design. A superpopulation description instead treats observed units as realizations from a probability distribution, typically with $\mathbf{X}_1, \dots, \mathbf{X}_n$ generated from a common law or from a structured dependence model. These perspectives answer different questions. A census may describe the realized finite population without sampling error, but it may still contain measurement error or missing values. Conversely, a probability sample can support generalization to a target population when its design and frame are understood, whereas a large nonprobability sample does not automatically do so.

A useful distinction is between *coverage*, *selection*, and *measurement*. Coverage concerns whether the sampling frame contains the target units. Selection concerns which frame units enter the sample and with what probabilities. Measurement concerns how the underlying attributes are translated into recorded values. If $\pi_i = \Pr(i \in s)$ denotes a unit's inclusion probability, inverse-probability weighting uses weights proportional to $1/\pi_i$ under conditions that make the design known and the relevant variables adequately measured. The corresponding estimator for a finite-population mean, for example, is a Horvitz–Thompson-type quantity

$$\hat{\mu}_{\text{HT}} = \frac{1}{N} \sum_{i \in s} \frac{Y_i}{\pi_i}.$$

Its interpretation depends on the sampling design, not simply on the number of records. When inclusion probabilities are unknown, or when selection depends on unobserved characteristics, design-based guarantees may not apply. Generalization then requires explicit assumptions, such as a model for selection or a transportability argument.

The familiar notation $\{(\mathbf{X}_i, Y_i)\}_{i=1}^n$ also conceals temporal ordering. In longitudinal data, $\mathbf{X}_{it}$ and Y_{it} may be indexed by unit and time, and an observation may represent a unit–occasion pair rather than a unit. In clustered data, let j index clusters and i members within cluster; then Y_{ij} may share a cluster-level disturbance. Treating all Y_{ij} as independent can make standard errors too small and can give a misleading impression of evidential strength. Before analysis, one should therefore record

the data-generating organization: who was eligible, when measurements were taken, how units were selected, and which observations share sources of dependence.

3.2 Variables, scales, and measurement

A variable is a rule assigning a recorded value to each unit, but the recorded value is not identical to the construct of interest. Let Z_i denote a latent or conceptual attribute, M_i the measurement process, and $X_i = M_i(Z_i)$ the observed value. The map M_i may introduce rounding, instrument noise, interviewer effects, recall error, classification error, or systematic differences across subgroups. Statistical analysis is consequently conditional on an operational definition of the variable. A measure of “income,” for example, must specify the reference period, currency, treatment of zero and negative values, and whether it is individual or household income.

Variable type describes the mathematical and substantive structure of a measurement. Binary variables take two categories, such as presence or absence of a condition. Nominal categorical variables have labels without an intrinsic order. Ordinal variables have an order, but the distances between adjacent categories are not necessarily equal. Interval scales support meaningful differences but not ratios, while ratio scales additionally possess a meaningful zero and permit ratio comparisons. Count variables take nonnegative integer values; continuous variables are idealizations whose possible values form an interval; and time-to-event variables require attention to censoring and the event definition. These distinctions constrain summaries and models. A mean of an ordinal response can be a useful descriptive convention in some settings, but it is not licensed merely by the presence of numerical codes.

Coding is a representation, not a substantive property. If education levels are coded 0, 1, 2, 3, replacing the labels by 10, 20, 30, 40 leaves their order unchanged but changes the arithmetic mean and variance. A nominal variable coded 1, 2, 3 should not be treated as quantitative merely because software accepts those values. Indicator notation can clarify intended comparisons: for categories $k = 1, \dots, K$, define $D_{ik} = \mathbb{I}(X_i = k)$. A contrast such as $\mathbb{E}[Y \mid X = 2] - \mathbb{E}[Y \mid X = 1]$ has a direct interpretation; a coefficient attached to the arbitrary code $X \in \{1, 2, 3\}$ generally does not.

Reliability and validity are distinct. Reliability concerns the stability or repeatability of a measurement, whereas validity concerns whether it represents the intended construct. A highly reliable proxy can be consistently wrong. For a continuous measurement with classical additive error, one may write $X_i = Z_i + U_i$, where $\mathbb{E}(U_i \mid Z_i) = 0$ and U_i is independent of Z_i . Even in this favorable case, regression coefficients involving X_i can be attenuated, and nonlinear functions such as ratios or logarithms can be distorted. Differential error, in which the distribution of U_i depends on outcome, exposure, or subgroup, may induce bias in either direction. Measurement quality is therefore part of the inferential argument rather than a preliminary cosmetic concern.

3.3 Data quality, coding, and analytic provenance

Data quality should be evaluated in relation to the estimand. A duplicate record, an implausible age, a unit mismatch, or an impossible date is not merely an inconvenience: each can alter the sample, the measurement scale, or the target of inference. Quality checks should include range and domain checks, consistency rules across variables, uniqueness of identifiers, temporal ordering, unit conventions, and the treatment of special codes such as “unknown,” “not applicable,” or “not asked.” A value of -99 may be numerically valid yet substantively missing. Conflating such codes with

genuine measurements contaminates means, correlations, regression fits, and graphical displays.

Let X_i^{raw} be a source value and $X_i^{\text{ana}} = g(X_i^{\text{raw}})$ the value used for analysis. The transformation g should be explicit. It may convert units, harmonize category labels, define a derived outcome, or winsorize extreme values. Each transformation changes the analysis and may change the estimand. For example, replacing a continuous biomarker by $\mathbb{I}(X \geq c)$ changes a question about average biomarker level into a question about prevalence above a threshold. It can also discard information and create sensitivity to the chosen cutoff. A principled analysis preserves the source variable when possible and reports the operational rules used to construct analytic variables.

Outliers require interpretation rather than automatic deletion. An extreme value may be a transcription error, a valid member of a heavy-tailed population, a measurement from a different process, or an influential but scientifically important case. Robust summaries, transformations, stratified displays, and sensitivity analyses can reveal whether conclusions depend on a small number of observations. Deletion is defensible only when a documented rule identifies an invalid observation or when the target population explicitly excludes it. Otherwise, deletion silently changes the population and can convert a data-quality decision into an unacknowledged substantive assumption.

3.4 Descriptive statistics as structured evidence

Descriptive statistics summarize the observed distribution; they do not by themselves establish a population claim or a causal relation. For a quantitative variable Y , the sample mean and variance are

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i, \quad s^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2.$$

The mean is sensitive to skewness and extreme observations, while the median and interquartile range describe central location and spread more robustly. The standard deviation measures dispersion around the mean; it is not a measure of uncertainty in the mean. Under independent, identically distributed sampling with finite variance, the standard error of $\bar{Y}$ is estimated by $s/\sqrt{n}$, but dependence, weighting, clustering, and finite-population corrections can change this calculation.

For a binary outcome $Y \in \{0, 1\}$, $\bar{Y}$ is the observed proportion of ones. For a categorical variable, counts and percentages should be accompanied by the denominator to which they refer. Percentages with different denominators are not directly comparable, and marginal percentages can obscure important differences in subgroup composition. Conditional summaries, such as $\widehat{\Pr}(Y = 1 \mid A = a)$, should identify both the conditioning set and the number of contributing observations. For skewed outcomes, reporting the arithmetic mean alone may conceal the distributional feature most relevant to decisions; quantiles, ranges, and domain-specific thresholds may be more informative.

Association summaries require similarly careful interpretation. The covariance is $\text{Cov}(X, Y) = \mathbb{E}[(X - \mathbb{E}X)(Y - \mathbb{E}Y)]$, and the Pearson correlation is

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X) \text{Var}(Y)}}.$$

Its sample analogue measures linear association on the selected scale. It is invariant to positive affine transformations but not to arbitrary nonlinear transformations, and it can be dominated by outliers. A zero correlation does not imply independence except under special distributional conditions, and a nonzero correlation does not identify a causal effect. Rank-based summaries may

be more appropriate for ordinal or markedly non-normal variables, but they answer a different question about ordering.

A descriptive table is strongest when it makes denominators, units, missingness, and subgroup definitions visible. A useful baseline table distinguishes the number of observed values from the nominal sample size and avoids presenting a long list of decimal places that exceeds measurement precision. Descriptive analysis should also be comparative: distributions may be shown overall and by design-relevant strata, while preserving the distinction between description and post hoc explanation. Exploratory analysis is valuable precisely because it can reveal unexpected structure, but patterns discovered after viewing the data should be treated as hypotheses unless confirmed by an appropriate design or independent evidence [4, 12].

3.5 Graphics and the geometry of distributions

A graph is a statistical argument about which aspects of the data deserve attention. Histograms and density displays reveal skewness, multimodality, gaps, and tails; empirical cumulative distribution functions show the proportion below every threshold; boxplots provide compact comparisons of location and spread; and scatterplots reveal nonlinearity, heteroscedasticity, clustering, and influential observations. For a binary or categorical variable, ordered bar charts with counts or percentages are generally more interpretable than encodings that imply quantitative distance.

Graphical choices should reflect the measurement scale and the comparison being made. Equal axes, common units, meaningful ordering, and visible denominators support accurate comparison. Overplotting can conceal concentration, while smoothing can create apparent features unsupported by the data. A fitted line may summarize a conditional mean but should not be mistaken for a structural law. Stratified displays can expose Simpson-type reversals in which a marginal association differs from within-stratum associations, although stratification itself does not determine which variables should be adjusted for. Principles of graphical perception emphasize that visual encodings differ in how accurately viewers can compare lengths, positions, areas, and colors [4]. The limitation is not that graphics are subjective; it is that every display selects a projection of a higher-dimensional data object.

Consider a treatment indicator A , an outcome Y , and a baseline risk group G . Suppose the overall treated and untreated means are 8 and 7, respectively, but within each risk group the treated mean is lower. The marginal difference can arise because treatment is more common in the high-risk group. A two-panel display of Y by A within levels of G can make the composition difference visible, while a single marginal plot cannot. The graph does not solve confounding, but it identifies the inferential question that must be addressed and discourages an interpretation based solely on the aggregate contrast.

3.6 Missingness and the observed-data problem

Missingness is part of the data-generating process. Let $R_{ij} = 1$ if variable X_{ij} is observed and $R_{ij} = 0$ otherwise. The observed data consist of $(R_{ij}, R_{ij}X_{ij})$, not the complete data matrix. Rubin’s framework distinguishes missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR) mechanisms [9, 10]. MCAR means that the missingness indicators are independent of both observed and unobserved values. MAR means, informally, that conditional on observed information, missingness does not depend on the missing values themselves. MNAR

denotes dependence that remains after conditioning on the observed data.

These are assumptions about a process, not properties that can generally be verified from the observed values alone. Under MCAR, complete-case analysis can preserve unbiasedness for some descriptive targets, but it may lose efficiency. Under MAR, complete cases can be systematically different from the full sample, and valid inference may require a likelihood or imputation procedure that incorporates predictors of missingness and of the incomplete variable. Under MNAR, additional assumptions, external information, or sensitivity analysis are needed. A missingness percentage is thus descriptive, whereas a claim that missingness is ignorable is inferential.

A simple example illustrates the issue. Let Y be an exam score and R indicate whether the score is recorded. If absent students are less likely to have scores recorded, then the observed mean $\bar{Y}_{R=1}$ estimates the mean among observed students, not necessarily the mean among all enrolled students. If attendance and prior performance are observed and sufficiently explain recording, adjustment may be plausible under MAR. If unobserved low performance still affects recording after adjustment, the target is sensitive to MNAR assumptions. Reporting only the complete-case mean hides this distinction.

Missingness can also be structural. “Not applicable” is not the same as an unobserved applicable value, and a value below a detection limit is not equivalent to an arbitrary blank. The analysis should distinguish these states and define the estimand accordingly. Sensitivity analyses may vary the assumed distribution of missing values, use bounds, or compare complete-case, weighted, and model-based results. The purpose is not to manufacture certainty but to show how much the conclusion depends on unverifiable assumptions.

3.7 The estimand: defining what is to be learned

An *estimand* is a precisely defined target quantity associated with a population, data-generating distribution, or causal system. It precedes the estimator, which is a function of the observed data, and the estimate, which is the value produced in a particular sample. Let P denote a population distribution. The prevalence of a binary characteristic is

$$\psi(P) = \mathbb{E}_P[Y] = \Pr_P(Y = 1),$$

while the difference in mean outcomes between exposure levels is

$$\psi(P) = \mathbb{E}_P[Y \mid A = 1] - \mathbb{E}_P[Y \mid A = 0].$$

The latter is an associational estimand unless A is assigned or additional causal assumptions are invoked.

Causal estimands require potential outcomes. Let $Y_i(1)$ and $Y_i(0)$ denote the outcomes that unit i would experience under treatment and control, respectively. The average treatment effect (ATE) is

$$\psi_{\text{ATE}} = \mathbb{E}[Y(1) - Y(0)],$$

where the expectation specifies the target population. The average treatment effect among the treated is

$$\psi_{\text{ATT}} = \mathbb{E}[Y(1) - Y(0) \mid A = 1].$$

These are distinct even when treatment is binary. Identification from observed data typically requires consistency, a well-defined intervention, exchangeability such as $(Y(1), Y(0)) \perp A \mid L$, and positivity

$0 < \Pr(A = a \mid L = l) < 1$ over relevant covariate values. Without these conditions, an estimated contrast may remain descriptive or model-dependent rather than a causal effect [7, 8].

Estimands also specify the outcome scale, time horizon, treatment version, and handling of competing events or censoring. A risk difference, risk ratio, odds ratio, mean difference, median difference, and restricted mean survival-time difference are not interchangeable. For a binary outcome, the risk difference is

$$\psi_{\text{RD}} = \Pr(Y(1) = 1) - \Pr(Y(0) = 1),$$

whereas the risk ratio is the quotient of the two probabilities. The odds ratio can be far from one when risks are common even if the risk ratio is moderate. Consequently, the choice of effect measure is a substantive decision about how evidence will be interpreted, not a choice to be made only after inspecting which result is most favorable [11, 13].

An estimand should state at least: the units and target population; the exposure, intervention, or comparison; the outcome and its measurement scale; the time point or follow-up window; and the summary functional. For example: “the two-year difference in mean systolic blood pressure under assignment to intervention versus usual care among eligible adults at baseline.” This statement rules out several ambiguities, including whether the target is the enrolled sample or a broader population, whether treatment received or treatment assigned is relevant, and whether deaths are assigned an outcome or treated as censoring.

3.8 Worked illustration: from records to a target quantity

Suppose a study enrolls $n = 400$ adults from two clinics to compare an intervention $A \in \{0, 1\}$ on a six-month symptom score Y , where lower scores indicate improvement. The observational unit is the enrolled adult, but clinic is a cluster and baseline symptom severity L is a pre-treatment variable. The study records age, clinic, assignment, follow-up score, and an indicator R for whether the six-month score is observed. The initial data description should report the recruitment frame, numbers assigned to each arm, clinic composition, baseline distributions, the score’s range and direction, and the amount and pattern of missing follow-up.

Assume that the observed arm means are $\bar{Y}_1 = 18.4$ and $\bar{Y}_0 = 20.1$. The raw difference -1.7 points is an estimator of an arm-specific associational contrast in the observed data. It is not automatically an unbiased estimator of the ATE. If assignment was randomized, the design may justify a causal interpretation, subject to adherence, attrition, interference, and the definition of the intervention. If assignment was observational, baseline severity and clinic may confound the comparison. If follow-up is missing more often among participants with poor outcomes, the complete-case difference targets a selected population unless missingness is handled under defensible assumptions.

A more explicit causal target is

$$\psi = \mathbb{E}\{Y(1) - Y(0) \mid S = 1\},$$

where $S = 1$ denotes eligibility for the study-defined target population. If the scientific question instead concerns all adults served by both clinics, the expectation must be taken over that service population, and the recruitment mechanism becomes relevant. If the target is the effect among those who would complete follow-up under either assignment, that principal-stratum target is different and generally not identified without stronger assumptions. Thus, the same records can support multiple analyses, but each analysis answers a different question.

The illustration also shows why exploratory analysis is not separable from inference. A plot of baseline severity by clinic may reveal imbalance or restricted overlap; a plot of follow-up score against baseline severity may suggest nonlinearity; and a missingness display may show that attrition differs by arm. These observations inform model adequacy and sensitivity analyses, but they do not license retrospective changes to the primary target without disclosure. Exploratory findings can motivate pre-specified or confirmatory analyses, while the distinction between discovery and confirmation protects the interpretation of uncertainty [2, 5].

3.9 Scope and limitations

No descriptive or inferential procedure can repair a population that was never defined, a measurement that does not represent its construct, or a selection process that excludes the units about which a claim is made. Larger samples reduce some forms of random error but do not necessarily reduce systematic selection, measurement, or missingness bias. Likewise, a narrower confidence interval reflects precision conditional on a model and design; it does not certify validity. Classical sampling theory provides a disciplined account of variation under specified assumptions [3, 14], but assumptions must be connected to the actual study.

The practical consequence is that every statistical result should be read as a statement indexed by its data, target population, measurement protocol, missingness assumptions, and estimand. Data description is therefore not an appendix to inference. It is the stage at which the objects of inference are made explicit, the permissible operations on variables are constrained, and the difference between an observed pattern and evidence about a broader or counterfactual quantity is established. Subsequent estimation and uncertainty quantification are meaningful only to the extent that this foundation is coherent.

3.10 Summary principles

A rigorous analysis begins by naming the observational unit and target population, distinguishing the sample from the population, and documenting selection and dependence. It then defines each variable’s construct, scale, coding, units, and quality limitations. Descriptive statistics and graphics should expose distributions, denominators, subgroup composition, and missingness rather than compressing them into opaque summaries. Finally, the estimand should be stated before selecting or interpreting an estimator: it specifies who is represented, what comparison is made, which outcome and time horizon matter, and what assumptions connect observed data to the desired quantity. These principles do not eliminate uncertainty, but they ensure that uncertainty is attached to a clearly articulated statistical question.

4 Sampling, Estimation, and Uncertainty

Statistical inference begins with a distinction between a data-generating process and the finite data set that is observed from it. The distinction is elementary, but it governs nearly every legitimate statement about evidence. A population, finite or conceptual, is represented through a probability model; a sample is a realization from that model; and an inferential procedure maps the realization to a claim about an unknown feature of the population. The resulting claim is necessarily conditional on assumptions about sampling, measurement, and model structure. Classical

mathematical statistics formalizes these ideas through random variables, sampling distributions, and procedures whose repeated-sampling properties can be studied before the observed data are available [3, 14]. Exploratory data analysis likewise emphasizes that inferential summaries should be interpreted in relation to the mechanism that produced the observations rather than as self-sufficient properties of a table [4, 12].

4.1 Random variables, populations, and samples

Let Y denote a random variable describing an outcome for a randomly selected unit from a target population. Its distribution, with cumulative distribution function F , may be discrete, continuous, or mixed. The expectation and variance, when they exist, are

$$\mu = \mathbb{E}_F(Y), \quad \sigma^2 = \text{Var}_F(Y) = \mathbb{E}_F\{(Y - \mu)^2\}. \quad (1)$$

More generally, a parameter $\theta = \theta(F)$ is a functional of the population distribution. The mean, median, variance, a regression coefficient, and a risk difference are all examples of parameters, although they answer different scientific questions. A parameter is fixed within a specified population model, whereas an observation is random before sampling. This distinction makes it inappropriate to describe an unknown parameter as “random” merely because the analyst does not know it. Randomness in the frequentist framework belongs to the sampling procedure and therefore to statistics computed from hypothetical repetitions of that procedure.

An i.i.d. random sample of size n from a superpopulation distribution F is a vector $Y_1, \dots, Y_n$ whose components are independent and identically distributed according to F . The i.i.d. assumption is a useful idealization, not a universal fact. Clustered observations, repeated measurements, spatial dependence, time series, informative nonresponse, and unequal selection probabilities alter the joint distribution. If the observations are dependent, the variance of an average is not generally σ^2/n ; for example, positive serial correlation inflates the variance relative to independent sampling. If selection is informative, even an infinitely large sample can converge to the wrong target. Consequently, uncertainty quantifies variation under a stated sampling design; it cannot repair a systematic mismatch between the observed sample and the population about which one wishes to speak.

For a random sample, the empirical distribution is

$$\hat{F}_n(y) = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(Y_i \leq y), \quad (2)$$

where $\mathbb{I}(\cdot)$ is the indicator function. The empirical distribution assigns mass $1/n$ to each observed value and is therefore itself a random object. It provides a nonparametric description of the sample, but it is not automatically identical to F . Its usefulness depends on the sampling assumptions and on the estimand: a sample median may be a suitable estimator of a population median, while a raw sample proportion may not estimate a population proportion if the sample was selected through a mechanism related to the outcome.

4.2 Sampling distributions and the logic of repetition

A statistic is a measurable function of the sample, $T = T(Y_1, \dots, Y_n)$. Before observing the data, T has a sampling distribution induced by the joint distribution of $(Y_1, \dots, Y_n)$. After observing $y_1, \dots, y_n$, the value $t = T(y_1, \dots, y_n)$ is fixed, but its inferential meaning is inherited from the

distribution of T under repeated samples. This counterfactual repetition is not a claim that the same physical study will literally be repeated; it is a mathematical device for defining calibration properties such as bias, coverage, and type-I error [3].

For the sample mean,

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i, \quad (3)$$

independence and finite variance give

$$\mathbb{E}(\bar{Y}) = \mu, \quad \text{Var}(\bar{Y}) = \frac{\sigma^2}{n}. \quad (4)$$

If $Y_i \sim \mathcal{N}(\mu, \sigma^2)$, then $\bar{Y} \sim \mathcal{N}(\mu, \sigma^2/n)$ exactly. Without normality, the central limit theorem states, under regularity conditions, that

$$\frac{\sqrt{n}(\bar{Y} - \mu)}{\sigma} \xrightarrow{d} \mathcal{N}(0, 1). \quad (5)$$

The theorem concerns the distribution of a standardized statistic as n increases. It does not say that the observations become normal, that a small sample is automatically safe, or that every statistic has a normal sampling distribution. Skewness, heavy tails, discreteness, dependence, and outliers can make finite-sample behavior materially different from its asymptotic approximation.

Asymptotic reasoning is valuable because exact finite-sample distributions are often unavailable. Let $\hat{\theta}_n$ estimate θ . Consistency means $\hat{\theta}_n \xrightarrow{P} \theta$, while asymptotic normality often takes the form

$$\sqrt{n}(\hat{\theta}_n - \theta) \xrightarrow{d} \mathcal{N}(0, V), \quad (6)$$

where V is an asymptotic variance. In regular parametric models, V may be estimated by a plug-in or sandwich estimator. These statements separate two questions that are frequently conflated: whether an estimator approaches the target and how rapidly its random error shrinks. A consistent estimator can have poor finite-sample performance, and an asymptotically efficient estimator can be misleading when its model assumptions are substantially violated.

4.3 Point estimators: bias, variance, and mean squared error

An estimator is a statistic designed to learn about a parameter. A point estimate is its realized value. For an estimator $\hat{\theta}$ of θ , the bias is

$$\text{Bias}_\theta(\hat{\theta}) = \mathbb{E}_\theta(\hat{\theta}) - \theta, \quad (7)$$

and the variance is $\text{Var}_\theta(\hat{\theta})$. Their combination is the mean squared error,

$$\text{MSE}_\theta(\hat{\theta}) = \mathbb{E}_\theta \left[(\hat{\theta} - \theta)^2 \right] = \text{Var}_\theta(\hat{\theta}) + \text{Bias}_\theta(\hat{\theta})^2. \quad (8)$$

An unbiased estimator has zero bias, but unbiasedness is not a universal criterion of superiority. A slightly biased estimator may have substantially lower variance and hence lower MSE. Conversely, a low estimated standard error does not establish that the estimator targets the scientifically relevant parameter.

For example, if $Y_1, \dots, Y_n$ have variance σ^2 , the estimator $\bar{Y}$ is unbiased for μ . The familiar sample variance

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2 \quad (9)$$

is unbiased for σ^2 under i.i.d. sampling with finite variance. Replacing $n-1$ by n produces the maximum-likelihood estimator under a normal model, but its finite-sample expectation is $(n-1)\sigma^2/n$; the difference is small for large n but conceptually important. Under a normal model, the exact independence of $\bar{Y}$ and S^2 supports the classical Student t interval. Outside that model, the same formula may be an approximation whose performance depends on sample size and distributional shape.

A useful illustrative calculation considers a population with $\mu = 10$ and $\sigma = 4$, sampled independently with $n = 16$. The sample mean has standard deviation $\sigma/\sqrt{n} = 1$. Thus, before observing the sample, a typical sampling fluctuation of the mean is on the order of one unit, not four units. If an analyst reports $\bar{y} = 12$, the difference from 10 is two standard errors under this hypothetical model. That calculation does not by itself establish that the population mean is 12, that the deviation is substantively important, or that the model is adequate. It only locates the observed estimate within a specified sampling distribution.

4.4 Standard errors and estimated precision

The standard error of an estimator is the square root of its sampling variance,

$$\text{SE}(\hat{\theta}) = \sqrt{\text{Var}(\hat{\theta})}. \quad (10)$$

Because the population variance is unknown, an estimated standard error, denoted $\widehat{\text{SE}}(\hat{\theta})$, is usually reported. For the mean under i.i.d. sampling,

$$\widehat{\text{SE}}(\bar{Y}) = \frac{S}{\sqrt{n}}. \quad (11)$$

The standard error describes sampling variability of an estimator, not variability among individual observations. A standard deviation of 10 and a standard error of 1 can coexist when n is 100. Confusing these quantities produces both exaggerated claims about individual-level predictability and understated claims about uncertainty in a population average.

In regression and generalized linear models, standard errors depend on the design matrix, error structure, and variance estimator. Heteroskedasticity-robust standard errors can address certain forms of nonconstant variance, while cluster-robust procedures address dependence within clusters, but neither automatically corrects omitted-variable bias, measurement error, selection bias, or a misspecified estimand. In prediction-oriented work, calibration, discrimination, and decision consequences may matter more than the standard error of an isolated coefficient [8, 11]. Measures such as net reclassification can also be sensitive to modeling and threshold choices, so their uncertainty and clinical meaning require careful qualification [13].

4.5 Confidence intervals and exact coverage

A confidence interval is a data-dependent procedure that returns an interval $C(Y_1, \dots, Y_n)$. Its frequentist coverage at parameter value θ is

$$\Pr_{\theta}\{\theta \in C(Y_1, \dots, Y_n)\}. \quad (12)$$

A nominal $(1 - \alpha)$ confidence procedure has exact coverage $(1 - \alpha)$ if this probability equals $(1 - \alpha)$ for every relevant θ , and approximate coverage if equality holds only approximately, typically in a large-sample limit. The correct interpretation is therefore a statement about the procedure across repeated samples: if the same sampling design and interval construction were repeated indefinitely under the assumed data-generating conditions, the stated proportion of intervals would contain the fixed parameter. It is not correct, in the frequentist interpretation, to say that the particular realized interval has a $(1 - \alpha)$ probability of containing the fixed parameter.

For a normal population with unknown variance, an exact interval for μ is

$$\bar{y} \pm t_{n-1, 1-\alpha/2} \frac{s}{\sqrt{n}}, \quad (13)$$

where $t_{n-1, 1-\alpha/2}$ is the corresponding Student t quantile. Exactness here is conditional on independent normal observations and the specified variance estimator. A large-sample Wald interval replaces the quantile by a normal quantile and is often written $\hat{\theta} \pm z_{1-\alpha/2} \text{SE}(\hat{\theta})$. Such intervals can be poorly calibrated for parameters near boundaries, for skewed estimators, and for small samples. For a binomial proportion, for instance, normal intervals can extend below zero or above one and can have actual coverage far from their nominal level; alternative constructions are preferable in many settings [1].

A numerical example clarifies the distinction. Suppose $n = 100$, $\bar{y} = 50$, and $s = 10$. A normal-quantile 95% interval is approximately 50 ± 1.96 , or $(48.04, 51.96)$. This interval is compatible with a range of population means under the model; it does not imply that 48.04 and 51.96 are equally plausible endpoints, nor that 95% of individuals lie inside the interval. A prediction interval for a future individual would be wider because it includes both uncertainty about the population mean and individual outcome variability. Interval width is thus tied to the estimand and the source of randomness.

Coverage is a property of the complete procedure, including sampling design, estimator, standard-error calculation, critical value, and any data-dependent selection. Reporting a nominal level without evaluating calibration can create false precision. Multiple interval reports, optional stopping, selective subgroup definitions, and post hoc transformation can all change the repeated-sampling behavior. The American Statistical Association’s principles emphasize that uncertainty should be communicated without treating a threshold such as $p = 0.05$ as a substitute for scientific reasoning [2, 15]. A confidence interval should therefore be read alongside the estimand, assumptions, design, missing-data strategy, and practical decision threshold.

4.6 Bootstrap and simulation-based uncertainty

When an analytic sampling distribution is difficult to derive, simulation can approximate it. A parametric simulation draws repeated samples from a fitted model $F_{\hat{\eta}}$, recomputes the estimator for each draw, and uses the resulting distribution to study bias, variance, tail behavior, or interval

coverage. This approach makes assumptions explicit: simulation from a misspecified model can provide a precise answer to the wrong question. It is most informative when the fitted model reproduces relevant dependence, heterogeneity, censoring, and selection mechanisms.

The nonparametric bootstrap treats the empirical distribution $\hat{F}_n$ as an approximation to F . For $b = 1, \dots, B$, draw a resample $Y_1^{*b}, \dots, Y_n^{*b}$ independently with replacement from the observed values, and calculate $\hat{\theta}^{*b}$. The bootstrap distribution of $\hat{\theta}^*$ approximates the sampling distribution of $\hat{\theta}$. Its standard-error estimate is

$$\widehat{\text{SE}}_{\text{boot}}(\hat{\theta}) = \left[\frac{1}{B-1} \sum_{b=1}^B \left(\hat{\theta}^{*b} - \bar{\hat{\theta}}^* \right)^2 \right]^{1/2}, \quad (14)$$

where $\bar{\hat{\theta}}^* = B^{-1} \sum_b \hat{\theta}^{*b}$. A percentile interval uses empirical quantiles of the bootstrap estimates; bias-corrected and accelerated intervals adjust for estimated bias and skewness. The bootstrap’s foundational account and subsequent treatment distinguish these constructions and their assumptions [6].

Consider estimating a sample median. Because the median’s sampling distribution can be skewed in small samples and is not generally summarized well by a symmetric normal approximation, a bootstrap distribution may reveal asymmetry directly. Yet resampling individual observations is not automatically valid for clustered or longitudinal data: the resampling unit must reflect the design, such as resampling clusters rather than rows. Similarly, a naive bootstrap can fail for extreme quantiles, nonsmooth statistics, sparse contingency tables, and estimators at parameter boundaries. Simulation and bootstrap are therefore tools for diagnosing and approximating uncertainty, not assumption-free replacements for probability theory.

4.7 Uncertainty beyond sampling variation

Sampling uncertainty is only one component of inferential uncertainty. Measurement error, missingness, nonresponse, treatment nonadherence, model uncertainty, and ambiguity in the target population can dominate the standard error. Rubin’s framework for missing data distinguishes mechanisms under which missingness may be ignored from those requiring explicit modeling or sensitivity analysis [9, 10]. If the probability of missingness depends on unobserved outcomes even after conditioning on recorded variables, conventional complete-case intervals need not have their advertised coverage.

A further distinction is between statistical uncertainty and decision uncertainty. An interval can be narrow while the estimand is poorly aligned with the decision, or wide while the plausible range is still useful for ruling out consequential effects. Bayesian posterior intervals use a different probability statement, conditional on a prior and likelihood; they should not be casually conflated with frequentist confidence intervals [7]. Both frameworks can be rigorous, but their probability meanings and calibration criteria differ.

The central discipline is to state what is random, what is fixed, which population is targeted, and which assumptions support the calculation. A point estimate answers “what value did the procedure produce?” A standard error addresses “how much would that procedure vary under repeated samples?” An interval addresses “which values are retained by a calibrated procedure under the stated model and design?” None of these questions alone establishes causality, practical importance, transportability, or truth. Good inference makes those boundaries visible rather than

hiding them behind additional decimal places. Exploratory summaries, formal estimates, and uncertainty intervals should consequently be treated as complementary representations of evidence, with their limitations documented as part of the result rather than relegated to an afterthought [2, 5].

4.8 Summary of interpretive safeguards

Several safeguards follow from the preceding framework. First, define the estimand before selecting an estimator: a mean, median, risk difference, odds ratio, and predictive risk are not interchangeable. Second, identify the sampling and missingness mechanisms; increasing n does not remove bias generated by selection or measurement. Third, report the estimator together with a measure of uncertainty and the assumptions behind that measure. Fourth, distinguish exact, approximate, and model-based calibration. Fifth, interpret interval width in relation to a scientifically meaningful scale, not merely in relation to statistical significance. Finally, treat simulation, bootstrap diagnostics, and sensitivity analyses as evidence about robustness, while recognizing that robustness is always relative to the alternatives examined. These safeguards preserve the mathematical content of inference while preventing a formally correct calculation from being presented as stronger evidence than the design and assumptions warrant.

In short, sampling distributions provide the bridge between observed data and population claims. Estimators summarize that bridge through bias, variance, and error; standard errors and intervals quantify selected aspects of its instability; and asymptotic, bootstrap, or simulation arguments approximate the relevant distributions when exact calculations are unavailable. The resulting uncertainty statement is credible only when its probability mechanism, estimand, and scope are explicit.

5 Evidence, Comparisons, and Categorical Inference

Statistical inference is an organized way to reason from observed data to claims about a data-generating process. The central object is not a verdict attached to a dataset, but a calibrated account of what the data make more or less plausible under explicit assumptions. This distinction is especially important when comparing groups, treatments, or measurements, and when outcomes are categorical rather than numerical. A comparison may be statistically detectable but scientifically trivial; conversely, a substantively important effect may be estimated imprecisely. The inferential task is therefore to connect a parameter, an estimator, an uncertainty statement, and a decision or interpretation without confusing any one of these components with evidence itself [3, 14].

5.1 Hypotheses, estimands, and the meaning of a p -value

A hypothesis is a statement about the population distribution, a parameter, or a structural mechanism. Let Y denote an outcome and let X denote a treatment indicator. A two-group mean comparison can be expressed through the estimand

$$\Delta = \mathbb{E}(Y \mid X = 1) - \mathbb{E}(Y \mid X = 0),$$

whereas a comparison of event probabilities uses

$$\Delta_p = \Pr(Y = 1 \mid X = 1) - \Pr(Y = 1 \mid X = 0).$$

The null hypothesis H_0 specifies a reference value, often $\Delta = 0$, and the alternative H_1 specifies a departure from that value. A two-sided alternative $H_1 : \Delta \neq 0$ differs materially from a directional alternative such as $H_1 : \Delta > 0$. The estimand should be fixed before inspecting results, since changing the target after seeing the data converts a clear inferential question into an adaptive one.

A test statistic $T(Y)$ is constructed so that its sampling behavior is known or approximated under H_0 . The p -value is then

$$p = \Pr_{H_0}\{T(Y^{\text{rep}}) \text{ is at least as extreme as } T(y)\},$$

where “extreme” must be defined in relation to the chosen alternative and Y^{rep} denotes hypothetical repeated data generated under the null model. Thus, a p -value is a tail probability conditional on the null hypothesis and the test procedure. It is not the probability that H_0 is true, the probability that the result arose by chance, or the probability that the scientific effect is important. These interpretations are excluded by the conditioning: the calculation assumes a null model rather than assigning it a posterior probability [2, 3].

For example, suppose two independent groups have sample means $\bar{y}_1 = 12.4$ and $\bar{y}_0 = 10.9$, with an estimated standard error for their difference of 0.60. The observed statistic for testing $H_0 : \Delta = 0$ is $t = 1.5/0.60 = 2.50$. If a t reference distribution with approximately 40 degrees of freedom is credible, a two-sided p -value is about 0.016. The appropriate conclusion is that the observed contrast would be relatively unusual under the specified null and sampling model. It is not that there is a 98.4 percent probability of a positive effect. That latter statement requires a different inferential framework and, in particular, a prior distribution and a posterior calculation.

Statistical evidence must also be separated from the multiplicity and flexibility of the analysis. If m hypotheses are tested, the probability of at least one false rejection can increase substantially even when every null is true. Pre-specification, correction for multiplicity when a family of claims is interpreted jointly, and transparent reporting of all relevant analyses are therefore part of the meaning of a reported p -value. Selective emphasis on favorable contrasts or thresholds can produce an apparently persuasive but poorly calibrated body of evidence [2, 12].

5.2 Errors, power, and effect size

A test has two basic error types. A Type I error rejects H_0 when it is true; its probability, under the design and reference model, is controlled by the significance level α . A Type II error fails to reject H_0 when a specified alternative is true; its probability is conventionally denoted β , and power is $1 - \beta$. These are properties of a procedure over repeated samples, not labels attached to a single realized decision. In particular, setting $\alpha = 0.05$ does not imply that the probability of a false claim in the present study is 0.05 unless additional assumptions about prior hypotheses and the decision process are supplied.

Power depends on the effect size, sample size, outcome variability, design, and rejection rule. For a one-sample normal-mean test with known standard deviation σ , the standardized effect is $d = (\mu - \mu_0)/\sigma$, and increasing $|d|$ or the sample size raises power. A very large sample can make a minute departure from H_0 yield a small p -value, while a small study can fail to detect a large effect because its interval is wide. Consequently, inference should report an estimate and an uncertainty interval rather than a dichotomous significant/not-significant label. A confidence interval is a procedure that, over repeated samples, covers the fixed parameter at its nominal rate under its

assumptions; it is not strictly a probability statement about the parameter after the data have been observed.

Effect sizes should be expressed in units relevant to the scientific question. The raw difference $\hat{\Delta} = \bar{y}_1 - \bar{y}_0$ preserves the original scale. A standardized mean difference, such as

$$\hat{d} = \frac{\bar{y}_1 - \bar{y}_0}{s_p}, \quad s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_0 - 1)s_0^2}{n_1 + n_0 - 2},$$

can facilitate comparisons across measurement scales, but it may obscure practical meaning and depends on a defensible common standard deviation. For binary outcomes, the risk difference, risk ratio, and odds ratio answer different questions. No single effect-size summary is universally preferable; the estimand should be chosen in light of decision consequences, baseline risk, and the data-generating design [2, 5].

5.3 Independent comparisons and Welch inference

Consider independent observations Y_{1i} and Y_{0j} from two populations with means μ_1, μ_0 and variances σ_1^2, σ_0^2 . The natural point estimate is $\hat{\Delta} = \bar{Y}_1 - \bar{Y}_0$, with estimated standard error

$$\widehat{\text{SE}}(\hat{\Delta}) = \sqrt{\frac{s_1^2}{n_1} + \frac{s_0^2}{n_0}}.$$

Welch's t statistic is

$$t_W = \frac{\bar{Y}_1 - \bar{Y}_0 - \Delta_0}{\sqrt{s_1^2/n_1 + s_0^2/n_0}},$$

with approximate degrees of freedom

$$\nu_W = \frac{(s_1^2/n_1 + s_0^2/n_0)^2}{\frac{(s_1^2/n_1)^2}{n_1 - 1} + \frac{(s_0^2/n_0)^2}{n_0 - 1}}.$$

The method does not require equal population variances and is generally preferable to a pooled-variance test when group sizes and variances differ. Its validity still relies on independent sampling or assignment, a meaningful outcome scale, and a normal approximation that is adequate for the sample sizes and distributional shapes. Severe skewness, heavy tails, influential observations, or dependence can require robust, permutation, or model-based alternatives.

Suppose a treatment group has $n_1 = 25$, mean 12.4, and standard deviation 3.0, while a control group has $n_0 = 40$, mean 10.9, and standard deviation 1.8. The estimated difference is 1.5, and

$$\widehat{\text{SE}} = \sqrt{3.0^2/25 + 1.8^2/40} \approx 0.70.$$

Welch's statistic is approximately 2.14, with degrees of freedom determined by the preceding expression rather than by $n_1 + n_0 - 2$. A 95-percent interval has the form $1.5 \pm t_{.975, \nu_W}(0.70)$. Reporting this interval makes clear both the plausible magnitude of the contrast and the precision with which it has been measured. If the treatment was not randomly assigned, the contrast is an association conditional on the sampling and measurement process, not automatically a causal effect.

5.4 Paired comparisons and repeated measurements

Paired data arise when each observation in one condition is naturally linked to an observation in another condition: before-and-after measurements on the same person, matched units, or repeated readings from one specimen. Let $D_i = Y_{1i} - Y_{0i}$ be the within-pair difference. The estimand is $\mu_D = \mathbb{E}(D)$, estimated by $\bar{D}$, with standard error $s_D/\sqrt{n}$. A paired t statistic is

$$t = \frac{\bar{D} - \Delta_0}{s_D/\sqrt{n}}.$$

The pairing is useful when the two measurements within a pair are positively correlated, because the variance

$$\text{Var}(D) = \text{Var}(Y_1) + \text{Var}(Y_0) - 2 \text{Cov}(Y_1, Y_0)$$

can be substantially smaller than the variance of an unpaired contrast. Treating paired observations as independent discards this structure and can misstate uncertainty.

The paired analysis requires that the differences, not necessarily the raw measurements in each condition, be sufficiently compatible with the reference approximation. It also requires that the pairing be substantively justified and that the pairs be independent of one another. A paired example illustrates the distinction: if ten participants are measured before and after an intervention and their mean change is -4.2 units with $s_D = 5.0$, the standard error is $5/\sqrt{10} \approx 1.58$. The evidence concerns the distribution of individual changes. It does not establish that every participant improved, nor does it establish causation if intervention exposure was confounded with time, selection, or co-interventions.

5.5 More than two means: ANOVA and planned contrasts

For K groups, one-way analysis of variance decomposes total variability around the grand mean into between-group and within-group components. With group means $\bar{Y}_k$, sizes n_k , and grand mean $\bar{Y}_{..}$,

$$SS_B = \sum_{k=1}^K n_k (\bar{Y}_k - \bar{Y}_{..})^2, \quad SS_W = \sum_{k=1}^K \sum_{i=1}^{n_k} (Y_{ki} - \bar{Y}_k)^2.$$

The classical model assumes $Y_{ki} = \mu_k + \varepsilon_{ki}$, independent errors with common variance σ^2 , and an appropriate response scale. The omnibus statistic

$$F = \frac{SS_B/(K-1)}{SS_W/(N-K)}$$

tests $H_0 : \mu_1 = \dots = \mu_K$. A small p -value indicates that the group means are not all equal under the model, but it does not identify which groups differ or quantify the most relevant contrast.

Planned contrasts should be stated in terms of the scientific comparison, for example $L = \mu_3 - (\mu_1 + \mu_2)/2$. A confidence interval for L is more interpretable than an unstructured sequence of post hoc tests. When many pairwise comparisons are examined, simultaneous error control or a clearly defined multiplicity strategy is needed. If variances differ materially, a heteroscedastic method such as Welch's ANOVA and variance-robust pairwise comparisons may be more credible than the equal-variance decomposition. If observations are clustered, repeated, or otherwise dependent, a simple one-way ANOVA has an incorrect error structure unless the dependence is modeled.

ANOVA also supports effect-size summaries such as

$$\eta^2 = \frac{SS_B}{SS_B + SS_W},$$

which describes the observed fraction of sample variation associated with group membership under the decomposition. It is not a universal causal measure and can be biased upward in finite samples. In all cases, residual diagnostics, design information, and substantive scale should accompany the omnibus result [7, 8].

5.6 Contingency tables and conditional proportions

For two categorical variables X and Y , a contingency table records joint counts n_{ij} for levels $i = 1, \dots, I$ and $j = 1, \dots, J$. The corresponding conditional proportions are distinct depending on the denominator:

$$\widehat{\Pr}(Y = j \mid X = i) = \frac{n_{ij}}{n_{i+}}, \quad \widehat{\Pr}(X = i \mid Y = j) = \frac{n_{ij}}{n_{+j}}.$$

Confusing these quantities is a common source of incorrect interpretation. If 30 of 100 exposed individuals experience an event and 10 of 50 unexposed individuals experience it, the event risks are 0.30 and 0.20, respectively. The risk difference is 0.10, and the risk ratio is 1.5. These calculations use exposure-group denominators; percentages within event-status columns answer a different question.

Under independence, expected counts are

$$E_{ij} = \frac{n_{i+}n_{+j}}{n_{++}}.$$

Pearson's chi-square statistic is

$$X^2 = \sum_{i=1}^I \sum_{j=1}^J \frac{(n_{ij} - E_{ij})^2}{E_{ij}},$$

which is compared with a chi-square distribution having $(I - 1)(J - 1)$ degrees of freedom under regularity conditions. The test addresses association, not causation, and a significant result does not specify its direction or practical importance. Cell contributions, standardized residuals, conditional proportions, and effect sizes should be inspected to understand the pattern.

The approximation can be unreliable when expected counts are small, especially in sparse tables or small samples. Fisher's exact test conditions on the observed margins and calculates a tail probability from the hypergeometric distribution; it is particularly useful for a 2×2 table with small counts. Exactness does not make the test assumption-free: the sampling scheme, fixed-margin interpretation, and choice of two-sided tail remain consequential. Continuity corrections, likelihood-ratio statistics, or model-based methods may be alternatives, but the choice should be motivated rather than selected solely to obtain a preferred conclusion [1].

5.7 Odds ratios and interpretation

For a 2×2 table with event counts a, b, c, d arranged as exposed-event, exposed-non-event, unexposed-event, and unexposed-non-event, the odds ratio is

$$\widehat{\text{OR}} = \frac{a/b}{c/d} = \frac{ad}{bc}.$$

An odds ratio of one corresponds to equal odds, values above one indicate higher odds in the exposed group, and values below one indicate lower odds. The logarithm is convenient because, under suitable large-sample conditions,

$$\text{SE}\{\log(\widehat{\text{OR}})\} \approx \sqrt{1/a + 1/b + 1/c + 1/d},$$

so an interval can be constructed on the log scale and exponentiated.

Odds ratios are not risk ratios. In the preceding example, the odds ratio is $(30 \cdot 40)/(70 \cdot 10) \approx 1.71$, whereas the risk ratio is 1.5. When events are common, the odds ratio can look considerably farther from one than the corresponding risk ratio. Logistic regression naturally parameterizes conditional odds, but its coefficients should not be described as changes in probability without converting them at specified covariate values or distributions. Sparse cells can yield unstable or infinite estimates, and separation in a regression model may require penalization or exact methods. Effect interpretation should therefore include baseline risks and absolute differences whenever possible [1, 8].

5.8 Confounding, randomization, and the limits of comparison

A comparison is confounded when a third variable is associated with both the exposure and the outcome and is not merely a consequence of the exposure on the causal path being studied. Let X be exposure, Y outcome, and Z a covariate. The crude association can differ from the association within levels of Z because the groups have different covariate distributions. Standardization, stratification, regression adjustment, matching, and weighting are ways to target a covariate-adjusted estimand, but they depend on adequate measurement, positivity, a defensible model, and a clear causal question.

Randomization addresses confounding by making treatment assignment independent of potential outcomes under the randomization mechanism; covariate balance is consequently achieved in expectation over randomizations. Under a randomized design, the difference in observed group means can provide an unbiased estimate of the average effect of treatment assignment, subject to the randomization scheme and assumptions concerning attrition and interference; with noncompliance, its interpretation is generally intention-to-treat rather than the average effect of treatment received. Randomization does not eliminate sampling variation or measurement error, and it does not automatically justify subgroup claims not supported by the design. In observational data, adjustment cannot recover information about unmeasured confounders merely by adding more covariates. Causal language should therefore match the design and assumptions, not the smallness of a p -value [7, 10].

A useful diagnostic is to state the counterfactual estimand explicitly. If $Y(1)$ and $Y(0)$ are potential outcomes under treatment and control, the average treatment effect is $\mathbb{E}[Y(1) - Y(0)]$. The observed data reveal at most one potential outcome for each unit. Randomization supplies a design-based route to compare the two potential-outcome distributions; observational adjustment requires assumptions such as conditional exchangeability, consistency, and positivity. These assumptions are substantive claims and should be examined through sensitivity analyses rather than hidden behind a generic “adjusted” label.

5.9 Missing data and analysis choices

Missingness changes the target population and can change the estimand. Let $R = 1$ indicate that an outcome or covariate is observed. Missing completely at random (MCAR) means that R is

independent of observed and unobserved data. Missing at random (MAR) means that, conditional on observed variables, missingness is independent of the missing values. Missing not at random (MNAR) denotes dependence that remains after conditioning on observed information. These are assumptions about a joint distribution, not properties that can generally be verified from observed data alone [9].

Complete-case analysis is simple and can be unbiased under MCAR, and in some settings under stronger model conditions, but it may waste information and alter the population represented by the analysis. Single imputation, such as replacing missing values by a mean, typically understates uncertainty because it treats imputed values as known. Multiple imputation generates several completed datasets, analyzes each, and combines estimates and within- and between-imputation variance. A principled missing-data analysis should report the amount and pattern of missingness, the variables used to model observation, the assumed mechanism, and sensitivity to plausible departures from MAR.

For an illustrative comparison, suppose outcome measurements are missing more often among participants with poor early response. A complete-case treatment difference may overstate benefit if loss to follow-up is related to later outcomes. Conversely, imputing every missing outcome with the overall mean can attenuate differences while producing overly narrow intervals. Neither direction is guaranteed; the point is that the missingness mechanism is part of the evidence model. Sensitivity analyses can vary assumptions about unobserved outcomes, use bounds, pattern-mixture specifications, or selection models. The resulting range of conclusions is often more honest than a single analysis presented as assumption-free.

5.10 Synthesis: evidence as calibrated comparison

Evidence for a comparison is strongest when the estimand is explicit, the design supports the intended interpretation, the estimator matches the outcome and dependence structure, uncertainty is quantified, and limitations are visible. A complete report of a two-group analysis should ordinarily identify the group definitions, sample sizes, missing-data rule, point estimate, interval, test statistic or reference procedure, and the assumptions that connect the calculation to the claim. For categorical outcomes, it should additionally specify the conditioning direction, absolute risks, and whether an odds ratio or risk ratio is being reported.

The logic can be summarized by the chain

question $\longrightarrow$ estimand $\longrightarrow$ design and assumptions $\longrightarrow$ estimator $\longrightarrow$ uncertainty $\longrightarrow$ interpretation.

Breaking this chain at any point can make a formally correct calculation scientifically misleading. A p -value without an effect estimate does not show magnitude; an effect estimate without an interval does not show precision; a conditional association without a design argument does not show causation; and a reported percentage without its denominator may not even identify the comparison being made. Exploratory graphics and robust summaries can reveal structure, but confirmatory claims require attention to the analysis plan, multiplicity, and model adequacy [4, 6, 12].

Finally, statistical evidence is cumulative and revisable. Replication, external validation, calibration of predicted risks, and sensitivity to reasonable analytic choices matter more than any isolated threshold crossing. Decision-relevant interpretation should ask not only whether the data are inconsistent with a null model, but also what range of effects remains compatible with the data, how consequential those effects would be, and which assumptions carry the conclusion. This approach

preserves the distinction between uncertainty and ignorance while recognizing that every inferential result is conditional on a design, a model, and a measurement process.

6 Regression and Model Specification

Regression is a language for expressing how the distribution of an outcome changes with one or more predictors. Its central object is not, in the first instance, a line drawn through a cloud of points, but a conditional distribution, denoted $Y \mid X = x$. A regression model specifies selected features of this distribution as functions of x , while leaving other features to a stochastic error term or to a fully specified probability law. This distinction matters because a fitted regression equation is not itself an explanation of the data-generating process. It is a set of assumptions about conditional means, variances, dependence, and—depending on the model—the shape of the outcome distribution. Model specification is therefore inseparable from statistical reasoning: the analyst must decide what is being modeled, what is being held fixed, which comparisons are meaningful, and which sources of uncertainty remain unrepresented.

6.1 Conditional mean models

Let Y be an outcome and let $X = (X_1, \dots, X_p)^\top$ be a vector of predictors. The conditional mean function is

$$m(x) = \mathbb{E}(Y \mid X = x). \quad (15)$$

A conditional mean model asserts that $m(x)$ belongs, exactly or approximately, to a specified family $m(x; \beta)$. In a linear conditional mean model,

$$\mathbb{E}(Y \mid X = x) = \beta_0 + \sum_{j=1}^p \beta_j x_j = x^{*\top} \beta, \quad (16)$$

where $x^* = (1, x_1, \dots, x_p)^\top$. The word “linear” refers to linearity in the unknown coefficients, not necessarily to a visual straight line in every original predictor. For example, x and x^2 can both be included as predictors while retaining linearity in (β_1, β_2) .

Equation (16) describes the mean response at a covariate value. It does not assert that every observation lies on the fitted surface, nor that the conditional variance is constant. A useful decomposition is

$$Y_i = m(X_i) + \varepsilon_i, \quad \mathbb{E}(\varepsilon_i \mid X_i) = 0, \quad (17)$$

where the zero conditional-mean restriction makes $m(X_i)$ the conditional expectation. The residual variation may be large, heteroskedastic, dependent, skewed, or generated by omitted variables. Thus, a coefficient can be interpreted as a statement about a conditional mean without claiming that the entire conditional distribution has been correctly specified.

For observational data, the target population and sampling mechanism must be stated before the coefficients acquire a clear meaning. A regression fitted to a convenience sample estimates a relationship in that sample unless additional assumptions justify transport to a wider population. Likewise, an association between Y and X_j is conditional on the other included predictors, but conditioning does not automatically remove confounding. These issues motivate the distinction between descriptive, predictive, and causal interpretations. Classical statistical theory provides

conditions for estimation and inference under a sampling model; it does not turn an arbitrary regression coefficient into a causal effect [3, 14].

6.2 Simple and multiple linear regression

In simple linear regression, one predictor X is used to model a quantitative outcome:

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i, \quad i = 1, \dots, n. \quad (18)$$

The intercept β_0 is the expected outcome at $X = 0$, provided that $X = 0$ is meaningful and lies within a relevant range. The slope β_1 is the expected change in Y associated with a one-unit increase in X . A one-unit change should be translated into substantive units: a slope of 2.4 might mean 2.4 examination points per additional hour of study, or 2.4 millimeters of blood pressure per unit of exposure. The numerical coefficient is not interpretable without its scale, coding, and reference point.

Multiple linear regression extends the model to several predictors,

$$Y_i = \beta_0 + \beta_1 X_{i1} + \dots + \beta_p X_{ip} + \varepsilon_i. \quad (19)$$

Here β_j is the difference in expected outcome associated with a one-unit increase in X_j while the other included predictors are held at fixed values. The phrase “holding fixed” is mathematical shorthand for comparing points in the covariate space; it does not necessarily describe an intervention that can be performed without changing the other variables. In the presence of collinearity, such comparisons may also be weakly identified by the observed data, producing unstable estimates and wide intervals even when the overall fitted values are precise.

For illustration, suppose an educational outcome Y is measured in points, X_1 is weekly study time in hours, and X_2 indicates prior preparation. A fitted model might be

$$\hat{Y} = 61.0 + 2.1X_1 + 5.8X_2. \quad (20)$$

The model estimates that two students with the same preparation status but differing by one study hour have predicted mean scores differing by 2.1 points. Students with the same study time but different preparation status differ by 5.8 predicted points. These are conditional associations. If prior preparation affects study time, or if motivation affects both study time and score, neither coefficient alone is a controlled causal effect.

6.3 Least squares and the geometry of fitting

Given observations (Y_i, X_i) , let $\mathbf{y}$ be the $n \times 1$ outcome vector and let $\mathbf{X}$ be the $n \times (p + 1)$ design matrix whose first column is a vector of ones. The linear predictor is $\mathbf{X}\beta$, and the residual vector is $\mathbf{e} = \mathbf{y} - \mathbf{X}\beta$. Ordinary least squares chooses the coefficient vector minimizing the residual sum of squares,

$$\hat{\beta} = \arg \min_{b \in \mathbb{R}^{p+1}} S(b), \quad S(b) = (\mathbf{y} - \mathbf{X}b)^\top (\mathbf{y} - \mathbf{X}b). \quad (21)$$

Expanding the objective gives

$$S(b) = \mathbf{y}^\top \mathbf{y} - 2b^\top \mathbf{X}^\top \mathbf{y} + b^\top \mathbf{X}^\top \mathbf{X} b. \quad (22)$$

Differentiating with respect to b and setting the gradient equal to zero yields the normal equations,

$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}. \quad (23)$$

If $\mathbf{X}^\top \mathbf{X}$ is nonsingular, then

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}. \quad (24)$$

The fitted vector $\hat{\mathbf{y}} = \mathbf{X} \hat{\boldsymbol{\beta}}$ is the orthogonal projection of $\mathbf{y}$ onto the column space of $\mathbf{X}$. Consequently, the residuals are orthogonal to every included regressor, including the intercept when present: $\mathbf{X}^\top \hat{\mathbf{e}} = 0$. This algebraic property is not evidence that the model is substantively correct; it follows from the optimization criterion.

Under the standard Gauss–Markov conditions—correct conditional mean specification, exogeneity $\mathbb{E}(\varepsilon \mid \mathbf{X}) = 0$, finite second moments, and no exact multicollinearity—ordinary least squares is unbiased and has minimum variance among linear unbiased estimators under the relevant homoskedastic formulation. Normality of errors is not required for the coefficient estimator to exist or to be unbiased, but it is often invoked for exact small-sample t and F distributions. With nonconstant variance, conventional standard errors can be misleading; heteroskedasticity-robust or design-based procedures may be appropriate. With clustered, longitudinal, or spatial observations, the independence assumption must be replaced by a dependence structure and the uncertainty calculation must reflect that structure.

The least-squares criterion also clarifies the role of influential observations. Because residuals are squared, observations with unusually large discrepancies can have disproportionate impact, especially when they possess high leverage in predictor space. A fitted line can therefore be sensitive to a small number of cases without any arithmetic error. Diagnostics should examine residual patterns, leverage, influence, and the range over which the model is being used. Transformations, robust methods, or alternative likelihoods may address particular problems, but no diagnostic automatically selects the scientifically correct model [7, 8].

6.4 Categorical predictors, interactions, and nonlinearity

Categorical predictors enter a linear model through indicator variables. If a variable has K categories, one category is ordinarily selected as a reference and represented by the intercept. For a three-level factor with levels A , B , and C , define $D_{Bi} = 1$ when observation i is in category B and 0 otherwise, and define D_{Ci} analogously. The model

$$Y_i = \beta_0 + \beta_B D_{Bi} + \beta_C D_{Ci} + \varepsilon_i \quad (25)$$

has expected means β_0 for A , $\beta_0 + \beta_B$ for B , and $\beta_0 + \beta_C$ for C . The coefficients are contrasts against the reference category, not intrinsic properties of the labels. Changing the reference changes the coefficient representation but not the fitted group means or their pairwise differences.

An interaction allows the association of one predictor to depend on another. With a binary group indicator G and a continuous predictor X ,

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 G_i + \beta_3 X_i G_i + \varepsilon_i. \quad (26)$$

For $G = 0$, the slope is β_1 ; for $G = 1$, it is $\beta_1 + \beta_3$. Thus β_3 is a difference in slopes, while β_2 is the group difference at $X = 0$. Centering X at a meaningful value changes the interpretation of the main group coefficient without changing fitted values. Omitting the interaction imposes

parallel regression lines, an assumption that should be motivated rather than inherited from software defaults.

Nonlinearity can be represented by transformations, polynomial terms, splines, or other basis expansions. A quadratic conditional mean,

$$\mathbb{E}(Y \mid X = x) = \beta_0 + \beta_1 x + \beta_2 x^2, \quad (27)$$

has marginal slope $\beta_1 + 2\beta_2 x$, so the effect of a one-unit increase varies with the starting value. The coefficient β_2 should not be interpreted in isolation as a universal “curvature effect.” Polynomial terms can be highly correlated, and extrapolation beyond the observed range can be especially hazardous. Graphical summaries and subject-matter knowledge are often needed to distinguish a meaningful nonlinear pattern from noise or an artifact of the chosen basis [4, 12].

6.5 Adjustment and causal caution

Adjustment means conditioning, mathematically or through a model, on measured covariates. It can improve precision, account for design variables, or reduce confounding when the relevant assumptions hold. It can also induce bias. A covariate that is a consequence of the exposure, a mediator on the pathway to the outcome, or a collider influenced by both exposure and outcome should not be treated as an ordinary control merely because it is available. The correct adjustment set depends on the causal question and the assumed data-generating structure, not on a universal rule to include every measured variable.

Suppose A is an exposure, Y an outcome, and C a pre-exposure common cause of both. A regression of Y on A without C may combine the association of A with Y and the backdoor path through C . Including C can support a conditional comparison, but causal interpretation additionally requires consistency, appropriate treatment definition, exchangeability conditional on C , and adequate positivity. These assumptions are substantive and cannot be verified solely by a high R^2 , a small p -value, or a well-behaved residual plot. Potential-outcomes reasoning formalizes the distinction between observed conditional associations and counterfactual contrasts [7, 10].

The phrase “adjusted effect” is therefore ambiguous unless the estimand is specified. A coefficient may represent a conditional mean difference, a controlled direct effect under stronger assumptions, or merely a parameter in a working prediction model. Selection based on statistical significance can further distort interpretation through multiple comparisons, omitted-variable bias, and post-treatment conditioning. Causal claims should be tied to a design, temporal ordering, and an explicit identification argument rather than to regression notation alone.

6.6 Generalized linear models

Generalized linear models extend the linear predictor beyond normally distributed continuous outcomes. A generalized linear model has three components: a random component specifying the outcome distribution from an exponential family; a systematic component, $\eta_i = \mathbf{x}_i^{*\top} \beta$; and a link function g satisfying

$$g\{\mathbb{E}(Y_i \mid X_i = \mathbf{x}_i)\} = \eta_i. \quad (28)$$

The link determines the scale on which predictors combine additively. It need not be the original outcome scale. For a count outcome, a log link gives $\log\{\mathbb{E}(Y \mid X)\} = X^{*\top} \beta$; for a positive

continuous outcome, a gamma model with a log link can express multiplicative mean changes. The variance may be related to the mean through a family-specific dispersion structure, so the assumption that the variance is constant is not universal across regression models.

A GLM does not eliminate specification concerns. The analyst must choose a distribution, link, covariates, functional forms, exposure or offset terms where relevant, and a dependence structure. A Poisson model, for example, may be unsuitable when the variance substantially exceeds the mean or when observations are correlated. Alternatives can include quasi-likelihood, negative-binomial models, random-effects models, or other distributions, but each introduces its own assumptions. Good fit on one scale does not guarantee calibrated predictions or correct tail behavior on another.

6.7 Logistic regression, likelihood, and odds ratios

For a binary outcome $Y_i \in \{0, 1\}$, logistic regression models the conditional success probability $p_i = \Pr(Y_i = 1 \mid X_i)$ through the logit link:

$$\text{logit}(p_i) = \log\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \sum_{j=1}^p \beta_j X_{ij}. \quad (29)$$

Equivalently,

$$p_i = \frac{\exp(\eta_i)}{1 + \exp(\eta_i)}, \quad \eta_i = \beta_0 + \sum_{j=1}^p \beta_j X_{ij}. \quad (30)$$

The linearity assumption concerns log-odds, not probability. A constant coefficient can correspond to very different changes in probability depending on baseline risk. For a one-unit increase in X_j , holding other covariates fixed, $\exp(\beta_j)$ is the conditional odds ratio. It is the ratio of odds at $X_j + 1$ to odds at X_j , not generally a risk ratio or a probability difference.

For example, consider a model for an adverse event with

$$\text{logit}(p) = -2.20 + 0.50X, \quad (31)$$

where X is an exposure measured in suitable units. At $X = 0$, the odds are $\exp(-2.20)$ and the probability is approximately 0.10. At $X = 1$, the odds are multiplied by $\exp(0.50) \approx 1.65$, while the probability increases to approximately 0.15. At a higher baseline risk, the same odds ratio would yield a larger absolute probability difference; at a lower baseline risk, the absolute difference would be smaller. A scientifically useful report should therefore accompany odds ratios with predicted probabilities, risk differences, or marginal standardization when those scales answer the question more directly [1, 11].

The logistic model is commonly estimated by maximum likelihood. Assuming conditional independence and $Y_i \sim \text{Bernoulli}(p_i)$, the likelihood is

$$L(\beta) = \prod_{i=1}^n p_i^{y_i} (1-p_i)^{1-y_i}, \quad (32)$$

and the log-likelihood is

$$\ell(\beta) = \sum_{i=1}^n [y_i \log(p_i) + (1-y_i) \log(1-p_i)]. \quad (33)$$

The maximum-likelihood estimator maximizes $\ell(\beta)$, typically by solving score equations numerically. The likelihood summarizes how compatible each parameter value is with the observed data under the assumed model; it is not the probability that a parameter value is true. Standard errors commonly arise from the observed or expected information matrix, with large-sample approximations supporting Wald intervals and tests. Likelihood-ratio and score procedures provide alternatives whose behavior can be preferable in some finite samples.

Complete or quasi-complete separation occurs when a linear combination of predictors nearly or perfectly distinguishes the two outcome classes. In that situation, the likelihood can increase as some coefficient tends toward infinity, so ordinary maximum-likelihood estimates may not be finite. Sparse data, rare events, influential observations, and high-dimensional covariates can make asymptotic standard errors unreliable. Penalization, Bayesian regularization, profile likelihood, or a redesigned analysis may be appropriate, but the remedy should be reported because it changes the estimand or inferential procedure. Model discrimination, calibration, and clinical or practical utility are distinct properties; a high area under a receiver-operating-characteristic curve does not establish well-calibrated probabilities or causal validity [8, 11, 13].

6.8 Interpretation, uncertainty, and limitations

Regression coefficients are estimates, not exact features observed without error. A confidence interval describes the performance of a procedure under repeated sampling and its stated assumptions; it does not assign a posterior probability to a fixed parameter unless a Bayesian model is being used. A p -value measures compatibility with a null hypothesis under the model and sampling procedure, not the probability that the null is true or the practical importance of the association. Estimates should be reported with uncertainty intervals and translated into substantively meaningful contrasts. When many specifications, outcomes, subgroups, or transformations are explored, selective reporting can make nominal uncertainty intervals overconfident.

A useful interpretive sequence is to identify the outcome scale, the covariate contrast, the conditioning set, and the target population. In a linear model, state the expected mean difference in outcome units. In a logistic model, state whether the result is an odds ratio, a probability difference, or a standardized population contrast. For nonlinear models and interactions, calculate predictions at representative covariate values or average predictions over the target distribution rather than relying only on a single raw coefficient. Centering, scaling, and reference-category choices should be documented because they alter coefficient meanings even when they leave fitted observations unchanged.

Prediction and explanation should also be distinguished. A model can predict well while providing a poor causal explanation, particularly when proxies, mediators, or post-outcome variables are included. Conversely, a substantively informative causal model may have modest predictive accuracy because important individual-level causes remain unmeasured. External validation is essential when predictions will be transported to new settings, since calibration and outcome prevalence can change. Missing data, measurement error, selection, and dependence can affect both estimates and uncertainty; complete-case analysis is not automatically unbiased, and nondifferential measurement error does not guarantee negligible bias.

Finally, model adequacy is partly empirical and partly conceptual. Residual and calibration diagnostics can reveal departures from assumptions, but no finite dataset proves a model true. Sensitivity analyses should vary plausible functional forms, adjustment sets, error structures, and

missing-data assumptions. Transparent reporting of the estimand, data provenance, inclusion rules, model specification, uncertainty method, and limitations makes regression results more reproducible and less vulnerable to overinterpretation. The fundamental discipline is to regard a fitted model as a conditional argument: given the sampling process, variables, functional form, and probability assumptions, what evidence does the observed data provide about the stated target? That discipline applies equally to ordinary least squares, generalized linear models, and more elaborate extensions [2, 5, 6, 9].

7 Model Checking, Prediction, Validation, Reproducibility, and Statistical Judgement

Statistical inference is not completed when an estimator, interval, or test has been computed. It is completed only after the analyst has examined those aspects of the assumed data-generating structure that are empirically assessable, considered the plausibility of assumptions that are not empirically testable, evaluated whether the resulting claims are stable under reasonable perturbations, and judged whether the strength of the conclusion is commensurate with the evidence. This section treats model checking and validation as constitutive parts of inference rather than as optional post-processing. The distinction is important because a numerical answer can be exactly calculated and nevertheless answer a poorly posed question, rely on an implausible sampling model, or conceal a failure of transportability. Classical statistical theory supplies distributions of estimators conditional on assumptions; model checking asks whether those assumptions are credible enough for the intended inferential use ([3, 14]).

7.1 The logic of model checking

Let $Y = (Y_1, \dots, Y_n)$ denote observed responses, let X denote observed covariates, and let a fitted model be indexed by a parameter $\hat{\theta}$ in a family $\{P_\theta : \theta \in \Theta\}$. A model is not merely an equation for a conditional mean. It is a joint collection of claims about the response distribution, dependence structure, covariate measurement, missingness, sampling, and the relationship between the observed variables and the estimand. For example, the linear model

$$Y_i = \beta_0 + X_i^\top \beta + \varepsilon_i, \quad \mathbb{E}(\varepsilon_i \mid X_i) = 0, \quad \text{Var}(\varepsilon_i \mid X_i) = \sigma^2, \quad (34)$$

contains stronger commitments than the mean equation alone. Independence, constant conditional variance, an appropriate functional form, and an adequate account of relevant covariates may all matter, depending on whether the objective is prediction, estimation, or a hypothesis test.

A useful formal device is a discrepancy or test statistic $T(Y, X)$ designed to reveal a particular failure. In a frequentist residual analysis, one compares the observed value $T(y, x)$ with its behavior under the fitted model, often using simulated or resampled data. In a posterior predictive analysis, one draws replicated data Y^{rep} from the fitted predictive distribution and examines whether $T(Y^{\text{rep}}, X)$ resembles $T(y, x)$ ([7]). The point is not to certify a model as true. No finite diagnostic can do that. Rather, diagnostics identify departures that are relevant to the proposed use and thereby guide revision, sensitivity analysis, or a narrower conclusion.

Model checking should therefore be tied to a stated inferential target. A curvature pattern may be consequential for estimating a marginal slope but immaterial for ranking subjects by predicted

risk. Conversely, a model with excellent average fit may be unsafe for individual prediction if it is badly calibrated in a clinically important subgroup. Exploratory graphics and numerical summaries are complementary: visualization can expose structure not anticipated by a pre-specified statistic, whereas numerical summaries make comparisons and reporting more reproducible ([4, 12]).

7.2 Residual diagnostics and functional-form assessment

For a fitted regression model, a residual is a measure of the part of an observation not explained by the fitted systematic component. In ordinary least squares, the raw residual is $e_i = y_i - \hat{y}_i$, the standardized residual is $e_i/\hat{\sigma}$, and a studentized residual additionally accounts for the observation-specific variance induced by the fitted design. Residuals are not direct observations of the errors: they are correlated transformations of the data and may have reduced variance at high-leverage design points. Their interpretation must consequently be conditional on the fitted model and design.

Several residual displays serve distinct purposes. A plot of residuals against fitted values probes nonlinearity and nonconstant variance in the response scale. A plot against each continuous predictor probes omitted curvature or an incorrect transformation. Residuals ordered by time or spatial location probe dependence, while residuals against unmodeled grouping variables probe systematic heterogeneity. A smooth trend through a residual plot is evidence that the conditional mean specification may be incomplete, but the absence of a visible trend is not evidence that the model is correct. In particular, low statistical power, overplotting, and a narrow covariate range can conceal meaningful departures.

Consider a study relating an outcome to exposure x . Suppose the true conditional mean is

$$\mathbb{E}(Y \mid X = x) = \beta_0 + \beta_1 x + \beta_2 x^2, \quad (35)$$

while a linear fit omits x^2 . If the observed design is symmetric around zero, the estimated linear coefficient can approximate an average slope even though residuals display a U-shaped relationship with x . The correct interpretation is not simply that the coefficient is “biased” in every possible sense; it is that the linear coefficient is a projection parameter whose meaning depends on the covariate distribution. If the scientific question concerns a constant causal or descriptive slope, curvature is a substantive failure. If the question concerns an optimal linear summary over the observed range, the model may remain useful, provided the projection interpretation and uncertainty are reported.

Normal quantile plots are often used to assess error normality. Their value is greatest for detecting severe asymmetry, heavy tails, or outliers, not for deciding whether a plot is perfectly straight. In moderate and large samples, mild nonnormality may have little effect on a mean estimate but can materially affect tail probabilities, prediction intervals, or small-sample tests. Robust or transformed analyses, distribution-free resampling, and explicit sensitivity analyses are preferable to treating a binary normality decision as a substitute for statistical reasoning ([14]).

7.3 Heteroskedasticity and dependence

Heteroskedasticity occurs when

$$\text{Var}(Y_i \mid X_i = x_i) = \sigma_i^2 \quad (36)$$

varies with x_i . In a linear model, ordinary least squares may still estimate the conditional mean consistently under exogeneity, but the conventional homoskedastic standard error is generally invalid.

The consequences include confidence intervals with incorrect coverage and tests with distorted size. A funnel-shaped residual plot is a familiar visual symptom, although variance patterns can be more complex than a monotone widening or narrowing.

The remedy depends on the purpose. Heteroskedasticity-consistent covariance estimators target uncertainty for regression coefficients under weaker variance assumptions, whereas weighted least squares changes the fitting criterion and is justified only when the variance model or efficiency objective is credible. A transformation, such as modeling a logarithmic outcome, can alter both the variance and the estimand; retransformation to the original scale then requires care because $\mathbb{E}(\exp Y \mid X) \neq \exp\{\mathbb{E}(Y \mid X)\}$ in general. A prediction interval should reflect both parameter uncertainty and the conditional outcome variance, not merely the uncertainty in the fitted mean.

Dependence is a separate issue. Repeated measurements on the same individual, clustered observations, temporal series, and spatial data violate the premise that observations provide independent pieces of information. If a cluster-level effect is ignored, standard errors may be too small and validation splits may leak information across nominally separate sets. Let m be cluster size and let ρ be an intraclass correlation. In a stylized balanced design, the effective sample size is approximately $n/[1 + (m - 1)\rho]$, illustrating why a nominally large sample can contain comparatively little independent information. Checking dependence requires design knowledge, not only residual plots. The analyst should state the observational unit, the clustering structure, and whether uncertainty is clustered, modeled, or otherwise adjusted.

7.4 Leverage, influence, and outlying observations

An outlier is unusual in its response, covariates, or both; leverage concerns unusual covariate configurations; influence concerns the extent to which an observation changes a fitted result. These are related but nonidentical concepts. In linear regression with design matrix X , the hat matrix is

$$H = X(X^\top X)^{-1}X^\top, \quad (37)$$

and h_{ii} measures the leverage of observation i . High leverage is not automatically problematic: an observation can be scientifically valid and highly informative about a region of the covariate space. It becomes influential when its leverage combines with an atypical residual to materially alter estimates.

Influence diagnostics compare the full-data fit with a perturbed fit. For example, Cook's distance may be written schematically as

$$D_i = \frac{(\hat{\beta} - \hat{\beta}_{(-i)})^\top X^\top X (\hat{\beta} - \hat{\beta}_{(-i)})}{p\hat{\sigma}^2}, \quad (38)$$

where $\hat{\beta}_{(-i)}$ is obtained after deleting observation i and p is the number of fitted coefficients. Such measures are diagnostic scales, not universal rejection thresholds. An influential point may indicate a data error, a distinct population, an omitted interaction, a boundary case, or a genuinely informative observation.

A defensible analysis investigates the provenance and scientific meaning of influential observations, reports whether they were retained, and shows how key estimates change under justified alternatives. Deleting observations solely to make a result significant is not a valid robustness analysis. At the same time, retaining a demonstrably erroneous measurement without qualification is not objectivity.

The relevant question is whether the conclusion is conditional on a small number of observations and whether that dependence is scientifically acceptable. Sensitivity analysis should distinguish a primary analysis chosen in advance from exploratory analyses prompted by diagnostics.

7.5 Logistic regression, separation, and binary-outcome diagnostics

For a binary response $Y_i \in \{0, 1\}$, logistic regression specifies

$$\Pr(Y_i = 1 \mid X_i) = p_i, \quad \text{logit}(p_i) = \log\left(\frac{p_i}{1 - p_i}\right) = \beta_0 + X_i^\top \beta. \quad (39)$$

The coefficient β_j is a conditional log odds ratio for a one-unit change in X_j , holding the other included covariates fixed. This interpretation is not generally a marginal population effect, and odds ratios can appear large even when absolute risk differences are modest, especially when baseline risks vary.

Complete separation occurs when a linear combination of predictors perfectly partitions observed cases and noncases. Under complete separation, the ordinary maximum-likelihood estimate does not exist as a finite vector: the likelihood increases along a direction in which fitted probabilities approach zero or one. Quasi-separation produces a related instability. Warning signs include enormous coefficients, huge standard errors, fitted probabilities numerically near the boundary, and a sharp dependence on a few observations. Separation is not evidence of an infinitely strong scientific relationship; it is evidence that the observed design provides insufficient overlap for the conventional likelihood analysis.

Diagnostics for logistic models should examine residuals appropriate to the binomial likelihood, leverage and influence, the distribution of fitted probabilities, and calibration. Deviance residuals summarize each observation's contribution to twice the difference between a saturated and fitted log likelihood, while Pearson residuals compare observed and expected outcomes using the binomial variance. Neither residual type can recover information absent from the data. With rare outcomes, apparent fit may be dominated by a small number of events, and grouped goodness-of-fit summaries can depend strongly on arbitrary binning.

When separation or sparse data is present, penalized likelihood, Bayesian regularization, exact methods, or a scientifically motivated reduction of model complexity may be appropriate. The method must be stated because regularization changes the estimand or sampling procedure. A stabilized estimate is not automatically an unbiased estimate of a conventional maximum-likelihood parameter, and a narrow interval obtained by imposing strong prior information should not be described as if it arose from the data alone.

7.6 Prediction: calibration, discrimination, and decision relevance

Prediction concerns the distribution of a future or otherwise unobserved outcome conditional on available information. For binary outcomes, a predictor produces a risk score $\hat{p}_i$. Calibration concerns agreement between predicted probabilities and event frequencies. In ideal calibration,

$$\Pr(Y = 1 \mid \hat{p} = p) = p \quad (40)$$

for relevant values of p . Discrimination concerns ordering: whether cases tend to receive higher scores than noncases. The area under the receiver operating characteristic curve can be interpreted

as the probability that a randomly selected case receives a higher score than a randomly selected noncase, subject to handling ties.

Calibration and discrimination answer different questions. A model can rank individuals well but be systematically overconfident, for example predicting risks near 0.9 for a group whose event frequency is 0.6. Conversely, a model that predicts nearly the sample prevalence for everyone can be well calibrated on average while having little discrimination. Calibration should be examined with calibration plots, calibration intercept and slope, proper scoring rules such as the Brier score, and uncertainty bands. Discrimination should be accompanied by confidence intervals and, where relevant, precision-recall summaries for rare outcomes. A single threshold-dependent accuracy is usually inadequate because it depends on prevalence and the chosen decision threshold.

Prediction must also be evaluated in the population and time point for which it is intended. Spectrum differences, changing treatment patterns, measurement shifts, and missingness can degrade transportability. Decision-curve reasoning can connect predictive performance to consequences by examining net benefit over plausible thresholds ([13]). Even then, a favorable curve does not establish clinical utility without credible outcome measurement, an appropriate action pathway, and consideration of harms from false positives and false negatives.

7.7 Validation, overfitting, and resampling

Overfitting occurs when a fitted procedure adapts to idiosyncratic noise in the development data and consequently performs worse on new data. If a model is selected, tuned, or repeatedly inspected using the same observations on which performance is reported, the apparent error is optimistically biased. The distinction between training and evaluation is therefore conceptual before it is computational: the evaluation data must not contribute to choices whose performance it is used to measure.

A held-out test set estimates future performance only under exchangeability between the test observations and the target population. A single split can be unstable, especially with small samples or rare events. Cross-validation averages performance across held-out folds, but its validity depends on respecting dependence, time order, clustering, and all stages of model selection. If feature selection or tuning occurs before cross-validation rather than within each training fold, information from the validation outcomes leaks into the fitted procedure and produces optimistic estimates.

The bootstrap provides another form of resampling. Given empirical distribution $\hat{F}_n$, one draws samples from $\hat{F}_n$ and recomputes an estimator or performance measure. The bootstrap distribution approximates the sampling distribution under the empirical population, and can quantify uncertainty for complicated statistics ([6]). It is not a universal repair: ordinary case resampling is inappropriate for dependent observations unless the resampling scheme respects the dependence, and bootstrap behavior can be poor at boundaries, under extreme sparsity, or for nonregular estimators. For predictive validation, optimism correction and bootstrap validation may be useful, but the resampling algorithm, number of repetitions, performance metric, and aggregation rule must be reported.

An illustrative example clarifies the issue. Suppose $n = 100$ observations contain ten events and a researcher evaluates twenty candidate predictors, selecting the model with the highest in-sample AUC. Even if every predictor is independent of the outcome, the maximum among twenty noisy AUC estimates will tend to exceed 0.5. Reporting that maximum as if it were pre-specified evidence confounds search with confirmation. Nested validation, in which the inner loop selects the model and the outer loop evaluates the complete selection procedure, targets the performance of the actual modeling pipeline rather than the performance of an idealized final fit. For small samples,

transparent acknowledgment of imprecision may be more honest than elaborate validation with unstable folds.

7.8 Reproducible reporting and transparency

Reproducibility is a property of the inferential record, not merely the ability to regenerate a numerical table. A transparent report identifies the estimand, target population, observational unit, inclusion and exclusion rules, missing-data treatment, outcome definition, covariate transformations, model family, estimation method, uncertainty procedure, diagnostic decisions, and deviations from the original analysis plan. It distinguishes confirmatory analyses from exploratory analyses and reports the extent of multiplicity or model search. These practices operationalize the principle that statistical conclusions should communicate both what was learned and how much discretion entered the analysis ([2, 9]).

Missing data require particular care. Let R_i indicate whether a variable is observed. Complete-case analysis can preserve unbiasedness for some descriptive estimands under missing completely at random, though efficiency may be reduced; validity for regression parameters depends additionally on the analysis model and the missingness structure. It may remain valid in particular regression settings under further assumptions. Under missing at random, likelihood-based methods or multiple imputation can be justified when the observed-data model is correctly specified; under missing not at random, identification generally requires sensitivity parameters or substantive assumptions ([9, 10]). A report should state whether missingness was modeled, how imputed values were generated, whether the outcome entered the imputation model, and whether conclusions changed across plausible missingness scenarios.

Reproducible reporting also requires preserving the chain from raw data to claim while respecting privacy and access constraints. Data provenance, versioned definitions, a frozen analysis dataset or data extract, a record of exclusions, and machine-readable tables help another analyst determine whether the same estimand was actually evaluated. When data cannot be shared, synthetic examples, detailed schemas, audit trails, and independently checkable summary quantities can still make the analysis more inspectable. Transparency does not require disclosing confidential information; it requires disclosing consequential choices and their rationale.

7.9 From diagnostics to statistical judgement

Statistical judgement is the disciplined synthesis of model adequacy, uncertainty, design limitations, and substantive consequences. It is not the selection of a preferred p -value after inspecting alternatives. A useful conclusion answers four questions: what quantity was estimated, under what assumptions; how sensitive is it to plausible perturbations; what population and decision context does it address; and what evidence would change the conclusion? Effect sizes and intervals should be interpreted in relation to scientific or practical scales, not only to a null value. A narrow interval around a negligible effect can support a meaningful negative conclusion, whereas a wide interval crossing important values supports uncertainty rather than proof of no effect ([2, 5]).

Diagnostics should influence conclusions proportionately. A modest heteroskedasticity pattern may call for robust uncertainty rather than abandonment of the mean model. Severe separation may preclude an ordinary logistic odds-ratio claim but still permit a descriptive statement about observed overlap. An influential observation may warrant a conditional conclusion: “the estimated association

is positive in the full sample and remains positive after the pre-specified quality exclusion, but its magnitude is sensitive to two high-leverage observations.” Such language is stronger scientifically than either concealing the sensitivity or treating every diagnostic flag as an automatic disqualification.

The final inferential object is consequently a joint statement about data, model, and purpose. For estimation, one should emphasize identification, bias, variance, and the interpretation of the estimand. For prediction, one should emphasize out-of-sample calibration, discrimination, uncertainty, and transportability. For decision support, one should additionally state thresholds, utilities, and potential harms. Causal interpretations require still stronger design and identification assumptions than predictive associations; adjustment for a variable because it improves fit does not by itself establish confounding control. These distinctions are central to responsible statistical communication ([7, 8, 11]).

A worked concluding example illustrates the hierarchy. Suppose a fitted binary-outcome model reports an odds ratio of 1.8 for an exposure, with a nominal 95% interval (1.2, 2.7). Residual and influence checks reveal one high-leverage cluster; calibration is poor in an external subgroup; and the exposure effect shrinks to 1.3 under a cluster-aware analysis. The appropriate conclusion is not that the original interval proves a stable effect. A defensible conclusion is that the development data are compatible with a positive conditional association, but magnitude is sensitive to dependence and influential observations, while external calibration limits transportability. Further data with adequate subgroup representation would be needed for a reliable risk prediction or intervention claim. This conclusion preserves the signal without overstating evidence.

In this sense, model checking is not an adversarial ritual imposed after inference. It is the mechanism by which an analysis discovers the domain in which its claims are credible. Validation disciplines optimism; resampling quantifies instability; reproducible reporting exposes discretion; and statistical judgement connects mathematical results to the limits of design and measurement. The mature analysis therefore reports not merely an answer, but the assumptions, diagnostics, perturbations, and target population that make the answer interpretable.

8 Conclusion

The central educational claim of this paper is not that graphical interfaces solve the problem of statistical inference. They do not. Statistical conclusions remain conditional on a population, a data-generating process, a measurement system, an estimand, a model, and a set of decisions that must be made visible and defensible. Nor is the claim that every question can be reduced to an automated pathway. Questions about causal effects, complex dependence, high-dimensional modeling, nonignorable missingness, and transportability often require more design information and methodological expertise than any introductory workflow can supply.

The more limited claim is that programming should not be confused with statistical reasoning. Participants can learn to distinguish observations from independent units, samples from populations, estimators from estimands, standard errors from standard deviations, confidence intervals from posterior probabilities, p -values from effect sizes, association from causation, and predictive discrimination from probability calibration without first mastering a programming language. They can also learn that a numerical result is not self-interpreting: its scientific meaning depends on assumptions, diagnostics, sensitivity, and the transparency of the analysis record.

A responsible no-code environment must therefore be guarded rather than frictionless. It should

make it harder, not easier, to ignore missingness, dependence, sparse data, separation, model misspecification, influential observations, validation leakage, or the distinction between explanatory and predictive aims. It should treat method selection as a consequence of question and design, not as a button chosen from a menu. And it should retain a reproducible record of the data, decisions, model, uncertainty, and limitations that produced a result. Under these conditions, no-code instruction can serve a defensible role: not the automation of statistical judgment, but the disciplined opening of statistical reasoning to learners whose first obstacle need not be software syntax.

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